Proof. Theorem 2.1 gives $A^{(\omega)} = \lim_{m\to\infty} f^m(A) = \chi(A)$ in SOT. By Lemma .7 (composition in the normal calculus), $A^{(\omega+1)} = f(A^{(\omega)}) = f(\chi(A)) = (f \circ \chi)(A) = \chi(A)$, where the last equality uses Lemma .9. Inductively, $A^{(\omega+\beta)} = \chi(A)$ for all $\beta \geq 1$. Hence stabilization occurs by stage ω. □

Remark .11 (Fejér-type monotonicity at the scalar level). Define the "Busemann" functional $B_1(z) := (1-|z|^2)/|1-z|^2$ on $\mathbb{D}$. For any holomorphic self-map g with $g(1) = 1$, Julia–Wolff–Carathéodory yields $B_1(g(z)) \geq cB_1(z)$ with $c \in (0, 1]$; composing along the cycle $(p_1, \ldots, p_K)$ shows that the sequence $m \mapsto B_1(f^m(\lambda))$ is nondecreasing up to a fixed multiplicative constant, a Fejér-type monotonicity toward the boundary point 1. We do not need this quantitative form in the proofs above; Proposition .10 relies only on the idempotence $f \circ \chi = \chi$. (For context see the proof of Lemma 2.6, p. 5.)

C.2 Power-boundedness of $\Psi(A)$ under H^∞ bounds (Banach case)

We retain the hypotheses of Theorem 3.1: A is Ritt/sectorial on a reflexive Banach space with a bounded H^∞ calculus on the spectral domain; $\phi_k \in H^\infty$, $\sup|\phi_k| \leq 1$, $\phi_k(1) = 1$; and $\Psi := \phi_K \circ \cdots \circ \phi_1$.

Lemma .12 (Sufficient condition for power-boundedness)**.** *If $\sup_\lambda |\Psi(\lambda)| \leq 1$, then $\Psi(A)$ is power-bounded.*

Proof. The H^∞ calculus yields a constant $C \geq 1$ with $\|g(A)\| \leq C\|g\|_\infty$ for all $g \in H^\infty$ on the spectral domain. Then $\|(\Psi(A))^n\| = \|(\Psi^n)(A)\| \leq C\|\Psi^n\|_\infty \leq C$ for all $n \in \mathbb{N}$, hence $\sup_n \|(\Psi(A))^n\| < \infty$. This is precisely the power-boundedness used in Theorem 3.1(a,b). (See Theorem 3.1 statement and proof, p. 6–7.) □

C.3 Boundary separation ⇒ joint fixed-points

We record the boundary case of condition (1) used with Lemma 3.3.

Proposition .13 (Boundary separation)**.** *Assume each ϕ_k has the peripheral fixed-point property: if $|\lambda| = 1$ and $|\phi_k(\lambda)| = 1$ then $\phi_k(\lambda) = \lambda$. If λ lies on the spectral boundary (e.g., $|\lambda| = 1$ in the discrete case) and $\Psi(\lambda) = \lambda$, then $\phi_k(\lambda) = \lambda$ for every k. Consequently, condition (1) in Section 3 holds for all boundary λ, and Lemma 3.3 gives* $\operatorname{Ran} P = \operatorname{Fix}(\Psi(A)) = \bigcap_{k=1}^K \operatorname{Fix}(\phi_k(A))$.

Proof. Suppose $|\lambda| = 1$ and $\Psi(\lambda) = \lambda$. Write the forward images $w_0 := \lambda$, $w_1 := \phi_1(w_0)$, …, $w_K := \phi_K(w_{K-1}) = \Psi(\lambda) = \lambda$. Each ϕ_k is Schur (bounded by 1), so $|w_j| \leq 1$ for all j. Since $|w_0| = |w_K| = 1$, every step must preserve modulus 1; otherwise once $|w_j| < 1$, all subsequent images stay strictly inside the domain, contradicting $|w_K| = 1$. By the peripheral fixed-point property, $|w_{j-1}| = |w_j| = 1$ implies $\phi_j(w_{j-1}) = w_{j-1}$, hence $w_j = w_{j-1}$. Therefore $w_j = \lambda$ for all j and $\phi_k(\lambda) = \lambda$ for each k. The stated identification with Lemma 3.3 follows. (See Section 3 and Lemma 3.3, p. 7–8.) □

C.4 A normalized two-layer example with $\alpha^* = \omega$ (replacement for Example 4.1)

We present a corrected two-layer construction that attains the extremal countable stage while staying within the stated Schur/peripheral assumptions. This subsumes the behavior intended in Example 4.1 and matches the parametric family in Example 4.2 at $t = 1/2$.

Example .14 (Normalized two-layer cycle)**.** Fix $t \in (0,1)$. Let $p_1^{(t)}(z) = t + (1-t)z$, $p_2^{(t)}(z) = b_t(z) := \frac{z-t}{1-tz}$, so that $p_1^{(t)}, p_2^{(t)}$ are Schur, $p_1^{(t)}(1) = p_2^{(t)}(1) = 1$, and b_t is the disk automorphism with $b_t(t) = 0$, $b_t(1) = 1$. Set $f_t := p_2^{(t)} \circ p_1^{(t)}$. A direct computation gives $f_t(z) = \frac{z}{1+t-tz}$. Conjugating by $T(z) = \frac{z}{1-z}$ (which maps $\mathbb{D}$ to the right half-plane) yields $T \circ f_t \circ T^{-1}(w) = \mu w$, with $\mu = \frac{1}{1+t} \in (\frac{1}{2}, 1)$. Hence $f_t^m(z) = T^{-1}(\mu^m T(z)) = \frac{\mu^m z}{1-(1-\mu^m)z} \xrightarrow{m\to\infty} \begin{cases} 1, & z = 1, \\ 0, & z \in \mathbb{D}, z \neq 1. \end{cases}$ Therefore $\sup_m |f_t^m| \leq 1$ and, for any normal A with $\sigma(A) \subset \overline{\mathbb{D}}$, Theorem 2.1 and Lemma .6 give $f_t^m(A) \to \chi(A) = E_A(\{1\})$ in SOT. No finite iterate is a projection, so the first stabilization occurs at the limit stage ω, i.e., $\alpha^* = \omega$. Taking $t = 1/2$ recovers the specific rate $\mu = 2/3$ alluded to in Section 4. (See Examples 4.2–4.3, p. 8–9.)

Remark .15. This normalized example replaces Example 4.1 verbatim: it satisfies the Schur bound, fixes 1, admits a closed-form iterate, and realizes the extremal countable stage without further assumptions.

C.5 One-line verification for the 2-cycle in Example 4.5

With $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, and $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, one has $SAS = J$ and $SJS = A$; thus the alternating, non-commuting layer update cycles $A \mapsto SAS = J \mapsto SJS = A$ with period 2. (See Example 4.5, p. 10.)

C.6 Consequences and cross-references

Proposition .10 validates the abstract's claim "a Fejér-monotonicity argument yields an ordinal bound $\leq \omega$" in the normal case, with the concrete stabilization proof relying on Lemma 2.6 and Appendix A. Lemma .12 provides the promised short justification that power-boundedness of $\Psi(A)$ follows when $\sup |\Psi| \leq 1$, as used in Theorem 3.1. Proposition .13 records the boundary separation needed for the joint fixed-point identification in Lemma 3.3. Example .14 supplies a clean ω-stage example consistent with Section 4 and requires no additional hypotheses beyond those already stated. This appendix is self-contained within the manuscript's framework and closes the questions noted in Sections 2–4 without altering any main theorem or adding external assumptions.

Appendix D — Riesz-projection framework for Lemma 3.3 in the Banach setting (non-normal case)

This appendix supplies a self-contained replacement for the argument following Lemma 3.3 in Section 3 that avoids spectral measures, works in the holomorphic H^∞ functional-calculus setting, and makes the isolation assumptions explicit. It also identifies the Riesz projection of $\Psi(A)$ at 1 with the product of the layer-wise Riesz projections and proves the desired equality of fixed-point spaces. Throughout we keep the notation and standing assumptions of Section 3 (Ritt/sectorial A on a reflexive Banach space with a bounded H^∞ calculus; $\phi_k \in H^\infty$ with $\sup |\phi_k| \leq 1$, $\phi_k(1) = 1$; $\Psi := \phi_K \circ \cdots \circ \phi_1$), and we refer to equations and statements there; see especially Theorem 3.1 and (1) on pp. 6–8.

D.0. Additional hypotheses (explicit isolation and separation)

We impose the following explicit conditions, which are either already used for convergence or are natural localizations compatible with the examples:

(D1) **Isolation at 1 for each layer.** For every k, 1 is an isolated point of $\sigma(\phi_k(A))$. Equivalently, there exists $r_k > 0$ such that $\sigma(\phi_k(A)) \cap \{\zeta : 0 < |\zeta - 1| \le r_k\} = \emptyset$. (This ensures that the Riesz projection of $\phi_k(A)$ at 1 is well-defined.)

(D2) **Separation of fixed points.** The implication (1) in Section 3 holds on the spectral domain: $\forall \lambda \in \sigma(A), \Psi(\lambda) = \lambda \implies \phi_k(\lambda) = \lambda$ for all k. (Sufficient boundary conditions appear in Proposition .13 of Appendix C; this is the exact hypothesis used beneath Lemma 3.3 on pp. 7–8.)

(D3) **Power-boundedness of the layers.** Each $\phi_k(A)$ is power-bounded. This holds under the standing H^∞ calculus bound and $\sup |\phi_k| \le 1$ just as in Lemma .12 (Appendix C.2) for Ψ.

Remark .16. (i) (D1) is a localization at $\zeta = 1$ in the image spectrum of each layer; it is automatic if $\phi_k(\sigma(A)) \subset \overline{\mathbb{D}}$ and 1 is the only unimodular value attained near the boundary, as in the boundary-separation statement (Appendix C, Proposition .13). (ii) (D2) is exactly the "separation condition" (1) referenced in the remark after Theorem 3.1; Appendix C, Proposition .13 verifies it for boundary spectral points. (iii) (D3) follows from the same calculus estimate used in Section 3 and Appendix C.2.

D.1. Layerwise Riesz projections via holomorphic calculus

For each k, fix a small positively oriented circle $\Gamma_k := \{\zeta : |\zeta - 1| = r_k\}$ contained in the resolvent set of $\phi_k(A)$ except for the enclosed point $\zeta = 1$. Define the Riesz projection at 1 for $\phi_k(A)$ by $Q_k := \frac{1}{2\pi i} \int_{\Gamma_k} (\zeta I - \phi_k(A))^{-1} d\zeta$. Then Q_k is a bounded idempotent commuting with $\phi_k(A)$ and with A. Moreover, since $\phi_k(A)$ is a holomorphic functional of A, each integrand $(\zeta - \phi_k)^{-1}(A)$ lies in the H^∞ calculus of A, hence Q_k itself is a holomorphic functional of A and consequently all Q_k commute pairwise and commute with $\Psi(A)$. (Use the algebra-homomorphism property of the H^∞ calculus quoted in Section 3, p. 6–7.)

Let $X_{1,k} := \operatorname{Ran} Q_k$ and $X_{0,k} := \ker Q_k$. These are closed A-invariant subspaces with $X = X_{1,k} \dotplus X_{0,k}$, $\sigma(\phi_k(A)|_{X_{1,k}}) = \{1\}$, $1 \notin \sigma(\phi_k(A)|_{X_{0,k}})$. (These are standard properties of Riesz projections; we use them only at the point $\zeta = 1$.)

D.2. Semisimplicity at 1 for power-bounded layers

Lemma .17 (No Jordan blocks at 1 for power-bounded maps)**.** *If T is power-bounded and 1 is an isolated point of $\sigma(T)$, then the Riesz projection Q of T at 1 satisfies* $\operatorname{Ran} Q = \ker(I - T)$.

Proof. On $M := \operatorname{Ran} Q$ one has $\sigma(T|_M) = \{1\}$. If $T|_M$ were not the identity, its Jordan form would contain a nontrivial nilpotent $N \neq 0$ with $T|_M = I + N$, hence $\|(T|_M)^n\| \ge cn$ for some $c > 0$, contradicting power-boundedness. Thus $T|_M = I$ and $\operatorname{Ran} Q = \ker(I - T)$. Apply this with $T = \phi_k(A)$ using (D3). □

Consequently, for each k, $X_{1,k} = \ker(I - \phi_k(A))$, $X_{0,k} = \overline{\operatorname{Ran}(I - \phi_k(A))}$. (This matches the mean-ergodic picture in Theorem 3.1(a), p. 6–7.)